

COMPLETE HANDBOOK

Patient-Centric Healthcare 5.0

Seven complete cases, four demonstration workshops, eleven prompts and a medical-AI implementation toolkit

AI assists • Professionals decide • Patients remain the purpose

Exactly 60 minutes theory + 90 minutes hands-on workshop

Doctors × Pharmaceutical Professionals

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DIGITAL COMPANION

The accompanying Streamlit app provides interactive navigation, tool filtering, readiness checks, prompt downloads and assessment.

How to use this handbook

PURPOSE

Use this as the single teaching companion for the 60-minute theory, 90-minute laboratory, independent case preparation and post-programme adoption planning.

Audience	Use
Facilitator	Run the exact clock, introduce evidence labels, conduct specialised-tool demonstrations and protect the debrief.
Doctors	Practise patient communication, diagnostic support, medicine verification, workflow design and professional accountability.
Pharma professionals	Practise neutral medical information, PV/PQC escalation, MLR review, patient education, evidence and professional boundaries.
Participants	Use the prompts and worksheets with synthetic/approved data, then design one controlled 30-day experiment.

Non-negotiable safety contract

- Use only synthetic/de-identified examples or approved public information in training.
- Do not use an AI assistant to make a final diagnosis, prescribe, change a dose, reassure away a red flag or delay urgent care.
- Do not enter output into a record, send it to a patient or publish it externally without the named professional review.
- Open original evidence; verify citations, calculations, medicine facts, guidelines, labels and local protocols.
- Follow organisational privacy, clinical governance, MI, pharmacovigilance, product-quality and MLR procedures.

UNIVERSAL DISCUSSION WRAPPER

You are supporting a synthetic healthcare education exercise. You are not the treating clinician and must not replace professional judgement. Do not provide a final diagnosis, prescribe, change a dose, provide reassurance or instruct anyone to delay urgent care. Separate supplied facts, patient-reported information, professional observations, assumptions, unknowns and AI-generated hypotheses. Do not invent findings, medicines, laboratory results, citations or guidelines. Identify red flags and information requiring urgent professional review. Use only the information and approved sources supplied. End with a section titled 'Human decision and verification required'.

Programme architecture

Block	Time	Method	Output
Theory	60 min	Seven evidence-labelled cases and guided discussion	Patient-centric operating principles
Workshop launch	8 min	Access map and universal safety wrapper	Group roles and safe-input contract
Four stations	75 min	Specialised-tool demonstration + paired ChatGPT practice	Four reviewed workflow outputs
Synthesis	7 min	POPLE scorecard and 30-day commitment	Controlled adoption experiment

Theory clock — exactly 60 minutes

Clock	Module	Facilitator focus
00–05	Opening and patient-purpose contract	Identify where technology steals attention or delays safety.
05–12	Capabilities + P-A-T-I-E-N-T	Own projection, research, diagnostic support and continuity under seven safeguards.
12–24	Operational value and alert design	Serbia EHR and SepsisLab.
24–34	Trust and adoption	Qatar perceptions and ambient scribes.
34–46	Learning and future readiness	China teaching and Trinity foresight.
46–54	Explainability and due diligence	Explanation supports scrutiny but does not prove benefit.
54–60	Synthesis + India guardrails	Evidence labels, governance and transition to the lab.

EVIDENCE-LANGUAGE RULE

Speak the result with its study design and limitation. A survey is not effectiveness evidence; a prototype is not an outcome trial; a heatmap is not proof.

P-A-T-I-E-N-T operating system

Letter	Checkpoint	Observable evidence
P	Protect privacy	Minimum data; approved environment; consent/notice; retention and access documented
A	Assess accuracy	Facts, omissions, calculations, citations, red flags and subgroup performance checked
T	Trace sources	Prompt, version, source, search and

Letter	Checkpoint	Observable evidence
		reviewer records preserved
I	Inform + consent	AI role, intended use, uncertainty and choice explained where relevant
E	Examine equity	Language, access, disability, literacy and subgroup effects evaluated
N	Never replace judgement	Licensed/authorised professional retains the decision
T	Take responsibility	Named reviewer, release, escalation, audit and stop rule

Part I — Seven complete theory cases

CASE-METHOD INSTRUCTION

Read the management dilemma first. Require participants to identify the decision, evidence design, patient-value outcome, risk, accountable professional and stop rule before discussing brands.

CASE 1

Serbia EHR — EHR-embedded X-ray triage and no-show prediction

MANAGEMENT DILEMMA

Should the organisation scale the models across facilities now, run a controlled extension, or redesign the workflow first?

Case context

A Serbian health system is considering how to scale two AI capabilities embedded in its electronic workflow: chest X-ray classification support and appointment-attendance prediction. The leadership question is not simply whether the models perform well, but whether the surrounding workflow improves access and preserves professional accountability.

Case narrative

Health Center Nis and the MEDIS.NET environment illustrate an operational approach in which AI is placed inside everyday information systems rather than offered as a separate experimental interface.

The published case reports approximately 92% chest X-ray classification accuracy for the described task and a substantial reduction in appointment non-attendance when predictions were combined with staff action. These are local operational signals, not universal guarantees.

The patient-centric challenge is to prevent a risk score from becoming an invisible decision. A high predicted no-show probability could trigger supportive outreach, but it could also create stigma, deprioritisation or inequity if staff do not understand the intended use and override rules.

For imaging, the model can highlight or classify, but a radiologist remains responsible for the report. The management design must make the hand-off, audit trail and escalation route visible.

Key facts

- Operational case and local model validation published in 2025.
- MEDIS.NET described across 25 facilities.
- VGG19-based chest X-ray classification support.
- Appointment-attendance prediction required staff action to create operational value.
- External and local validation remain necessary before transfer to another setting.

Primary source: <https://www.mdpi.com/2073-431X/14/8/320>

Case 1 evidence and patient-centric analysis

Reported signal	Evidence label	Limitation	Human checkpoint
The article reports about 92% X-ray classification accuracy and a substantial reduction in no-shows when models and staff action were combined.	2025 operational case study and local model validation	Single health-system context; model performance and workflow results require external and local validation.	Radiologist signs the report; staff decide outreach or scheduling action.

Stakeholder map

- Patients needing timely imaging or appointments
- Radiologists and referring clinicians
- Scheduling and outreach teams
- Health-system leadership, IT, privacy and quality teams
- Communities that may be differently affected by outreach rules

Patient-centric risks

- False reassurance or missed abnormality if classification is treated as a report.
- Unfair labelling or reduced access for patients predicted to miss appointments.
- Outreach that reveals sensitive information or ignores preferred communication channels.
- Performance drift when the model is transferred to another population or workflow.

CENTRAL LESSON

Embed probability in the workflow; do not hide decision ownership.

Case 1 teaching and transfer guide

Discussion question	Suggested direction
Which decision must remain visibly human in the imaging and attendance workflows?	Scale through a controlled extension, not an automatic enterprise rollout.
What should staff do—and never do—after a high no-show-risk prediction?	Use attendance risk for supportive outreach, never punitive scheduling or denial of access.
Which outcomes would demonstrate patient value beyond model accuracy?	Measure completed appointments, time-to-care, false alerts, outreach burden, patient preference, subgroup performance and complaints.
What subgroup and equity checks are required before expansion?	Retain radiologist sign-off and make model output, confidence, override and final report auditable.
What error or workflow failure should pause the pilot?	Define stop criteria for clinically meaningful misses, inequity, privacy incidents and unowned alerts.

Workshop bridge

Demo	Tools / prompt	Expected output	Human checkpoint
Redesign a prediction-to-action workflow	ChatGPT Prompt 8 plus an approved vendor evidence pack • Prompt 8	Vendor/workflow due-diligence table and controlled-pilot decision.	Clinical, privacy, equity, IT/security and operational owners approve before deployment.

- Define intended user, patient population and exact decision.
- Separate reported performance from evidence still required locally.
- Write supportive outreach rules, override and audit requirements.
- Choose patient-value, safety, equity and workload measures.
- Set continue, redesign and stop thresholds.

TRANSFER ASSIGNMENT

Draft a one-page pilot charter for appointment outreach that makes patient choice, staff responsibility, equity monitoring and stop criteria explicit.

Participant decision and evidence:

CASE 2

Qatar perceptions — What family physicians expect and fear

MANAGEMENT DILEMMA

How should a primary-care organisation introduce AI when clinicians see potential value but remain concerned about confidentiality, accountability and patient satisfaction?

Case context

Family physicians in Qatar were surveyed about their awareness and perceptions of AI applications in primary care. The case reveals that adoption depends on trust, accountability, confidentiality and patient experience—not awareness alone.

Case narrative

The cross-sectional survey received 102 eligible responses from 724 invited family physicians. It reported that 78.4% were aware of AI and that respondents perceived possible value in clinical, administrative and public-health tasks.

The same respondents raised concerns about confidentiality, accountability and patient satisfaction. Because this was a perception survey, it cannot establish clinical effectiveness or safety.

Leadership therefore faces a social and operating-model problem. A product demonstration may increase curiosity but will not answer who remains accountable, what data may be entered, how a patient is informed, or what happens when the tool is wrong.

A patient-centric introduction should begin with a bounded workflow, visible safeguards, training and a channel for clinicians and patients to report concerns.

Key facts

- 2024 cross-sectional perception survey.
- 102 eligible responses from 724 invited physicians.
- 78.4% reported AI awareness.
- Positive expectations coexisted with confidentiality, accountability and satisfaction concerns.
- The study did not test clinical effectiveness.

Primary source: <https://ai.jmir.org/2024/1/e40781>

Case 2 evidence and patient-centric analysis

Reported signal	Evidence label	Limitation	Human checkpoint
78.4% reported AI awareness; respondents perceived value in task management but raised confidentiality, accountability and patient-satisfaction concerns.	2024 cross-sectional perception survey	Self-reported perceptions from 102 of 724 invited physicians; not a clinical-effectiveness study.	Physicians asked for training, governance and clear accountability.

Stakeholder map

- Family physicians and clinical leaders
- Patients and caregivers
- Privacy, information-security and legal teams
- Training and quality-improvement teams
- Technology vendors and service owners

Patient-centric risks

- Patients may not understand that AI assisted the encounter or communication.
- Clinicians may over-trust or avoid a tool because accountability is unclear.
- Confidential information may enter an unapproved environment.
- Implementation may optimise task speed while reducing conversation quality.

CENTRAL LESSON

Adoption is a trust and operating-model problem, not only a model problem.

Case 2 teaching and transfer guide

Discussion question	Suggested direction
What assurance would clinicians need before using a new AI tool?	Select a low-risk, reviewable workflow such as synthetic training or draft patient education from approved evidence.
What notice, consent or explanation should a patient receive?	Publish a simple responsibility map covering input, review, release, escalation and audit.
Who owns an incorrect output and its downstream consequences?	Train users on limitations, privacy and verification rather than interface navigation alone.
Which first use case is valuable, bounded and reversible?	Collect both patient and clinician feedback; include opt-out and complaint routes where appropriate.
How should patient satisfaction and clinician trust be measured?	Treat trust as an outcome to measure, not a communication problem to manage away.

Workshop bridge

Demo	Tools / prompt	Expected output	Human checkpoint
Build an AI adoption assurance plan	ChatGPT Prompt 6 and the Demo Readiness Checklist • Prompt 6	Six-week controlled pilot charter.	Clinical governance, privacy, IT/security and service ownership approve the pilot.

- Choose one bounded primary-care workflow.
- Define intended and excluded uses.
- Map notice/consent, minimum data and accountable reviewers.
- Specify trust, experience, time and safety measures.
- Write rollback and stop criteria.

TRANSFER ASSIGNMENT

Create a one-page assurance statement that a clinician could explain to a patient and a regulator.

Participant decision and evidence:

CASE 3

China teaching — Generative AI in case-based medical learning

MANAGEMENT DILEMMA

How can faculty use generative AI to reduce unproductive search time without removing productive struggle and independent reasoning?

Case context

A comparative medical-education study at Chengdu Medical College examined the use of Kimi Chat 2.0 during a liver-cancer case assignment. It suggests efficiency and performance gains while exposing risks to reasoning, originality and interaction.

Case narrative

The study involved 79 students in two intact classes: a control group of 40 and an AI group of 39. The AI group used Kimi Chat 2.0 to research and present a liver-cancer case.

Reported assignment time was 2.6 hours for the AI group versus 5.5 hours for the control group. Final scores were 77.3 ± 4.3 versus 66.5 ± 5.4 , case scores 8.7 ± 2.1 versus 6.4 ± 3.5 , and teacher questions fell from 35 to 10.

These findings come from one course and two intact classes; they are educational signals, not clinical outcomes or universal proof that AI improves learning.

The patient-centric connection is indirect but important: medical education should produce professionals who can notice uncertainty, challenge assumptions and communicate responsibly. Instant answers can weaken those capabilities if the activity is not redesigned.

Key facts

- 2025 comparative educational study, n=79.
- Control n=40; AI group n=39.
- Kimi Chat 2.0 used for a liver-cancer case assignment.
- Reported reductions in assignment time and teacher questions, with higher scores.
- Single-course, non-randomised institutional context limits generalisation.

Primary source: <https://link.springer.com/article/10.1186/s12909-025-07567-z>

Case 3 evidence and patient-centric analysis

Reported signal	Evidence label	Limitation	Human checkpoint
Assignment time was 2.6 vs 5.5 hours; final scores 77.3±4.3 vs 66.5±5.4; teacher questions fell from 35 to 10.	2025 comparative educational study	Two intact classes in one course; not randomised across institutions; learning and grading findings should not be treated as clinical outcomes.	Faculty discussion remains essential for reasoning, originality and challenge.

Stakeholder map

- Students and faculty
- Curriculum and assessment leaders
- Future patients relying on graduates' reasoning
- Academic-integrity and technology teams
- Accreditation and quality-assurance bodies

Patient-centric risks

- Students may learn polished answer patterns without building diagnostic reasoning.
- Homogeneous AI outputs may reduce originality and alternative viewpoints.
- Fabricated or misapplied evidence may be rewarded when only final presentation quality is graded.
- Reduced faculty dialogue may remove opportunities to expose unsafe assumptions.

CENTRAL LESSON

AI raises the floor and compresses search time; it can also homogenise thinking.

Case 3 teaching and transfer guide

Discussion question	Suggested direction
What part of the assignment should be completed before AI is used?	Require an initial problem representation, hypotheses and uncertainty statement before AI access.
How can assessment reward revision and source verification rather than polish?	Use AI as a Socratic challenger that asks for evidence and contradiction rather than revealing the answer immediately.
What evidence would show that reasoning—not only marks—improved?	Grade source traceability, reasoning change, reflection and oral defence.
When does saved time become lost learning?	Use different cases or constraints to reduce look-alike submissions.
How should AI assistance be disclosed?	Faculty remain responsible for verified content and productive dialogue.

Workshop bridge

Demo	Tools / prompt	Expected output	Human checkpoint
Socratic case-based learning	ChatGPT Prompt 9 • Prompt 9	Staged Socratic dialogue, revised reasoning and rubric feedback.	Faculty verifies clinical content and preserves learner effort, originality and dialogue.

- Learner writes a problem representation before prompting.
- AI asks for common, dangerous and alternative hypotheses.
- Learner identifies evidence for/against and missing information.
- AI challenges one assumption and introduces one verified source.
- Learner revises; faculty assesses reasoning with a transparent rubric.

TRANSFER ASSIGNMENT

Redesign one existing medical or pharma training activity so the learner must commit, challenge, verify and revise.

Participant decision and evidence:

CASE 4

Trinity foresight — AI-supported scenario planning for medical curricula

MANAGEMENT DILEMMA

How can an institution use AI-generated futures to support strategy without allowing the tool's framing to determine institutional values?

Case context

A Trinity College Dublin workshop used generative tools to make four possible 2035 medical-education futures tangible. Participants then identified capabilities worth preserving across scenarios.

Case narrative

The 2024 participatory workshop involved 28 participants. ChatGPT supported scenario narratives, MidJourney produced visuals and Murf.AI supplied narration. Facilitators reviewed and refined all AI outputs.

Participants worked with four scenarios and converged on capabilities including systems thinking, patient-centredness, continuous learning, participatory health, sustainability and technological adaptability.

The study was qualitative and single-institution. It included no patient representatives and may have been affected by groupthink or framing. It should not be read as a universally applicable curriculum model.

The central lesson is that AI can increase the vividness and speed of foresight, while people must choose the values, trade-offs and actions. Patient participation should be designed into future exercises rather than assumed.

Key facts

- One participatory workshop with 28 participants.
- Four 2035 scenarios.
- ChatGPT, MidJourney and Murf.AI used to create reviewed scenario assets.
- Participants identified cross-scenario capabilities.
- No patient representatives and single-institution design limit transferability.

Primary source: <https://www.jmir.org/2025/1/e73173>

Case 4 evidence and patient-centric analysis

Reported signal	Evidence label	Limitation	Human checkpoint
Participants converged on systems thinking, patient-centredness, continuous learning, participatory health, sustainability and technological adaptability.	2025 single-institution qualitative participatory-design case	One workshop, no patient representatives, possible groupthink and framing effects; not universally generalisable.	Facilitators reviewed and refined all AI outputs; participants chose competencies and actions.

Stakeholder map

- Students, faculty and curriculum leaders
- Patients and community representatives
- Health-system and pharma partners
- Professional and accreditation bodies
- Technology, sustainability and ethics experts

Patient-centric risks

- AI framing may privilege technology-led futures over community priorities.
- Patients may be discussed as beneficiaries without participating in design.
- Compelling narratives and visuals can disguise weak assumptions.
- Consensus may reflect groupthink rather than robust disagreement.

CENTRAL LESSON

AI can make futures tangible; people must choose values and priorities.

Case 4 teaching and transfer guide

Discussion question	Suggested direction
Which capability remains essential in every plausible future?	Use multiple scenario prompts and deliberately include adverse or low-technology futures.
Whose voice is absent from the scenario process?	Invite patients/community representatives as co-designers with decision rights.
Which assumptions came from evidence and which came from AI framing?	Label AI-generated assumptions and require evidence or rationale for each.
What scenario would make today's strategy unsafe or irrelevant?	Use anonymous challenge rounds before convergence to reduce groupthink.
How should patient representatives participate?	Translate capabilities into measurable curriculum or workforce actions.

Workshop bridge

Demo	Tools / prompt	Expected output	Human checkpoint
Create and challenge a 2035 patient-care scenario	ChatGPT plus Canva or Gamma • Prompt 10	Reviewed scenario card and no-regret action.	Participants—not the model—choose values, priorities and commitments.

- Generate one optimistic, one constrained and one adverse scenario.
- Mark every assumption as evidence-based, value choice or speculation.
- Add a patient/community perspective and an equity consequence.
- Identify capabilities robust across all scenarios.
- Choose one no-regret action for the next 12 months.

TRANSFER ASSIGNMENT

Build a one-page future capability map that includes a patient representative's decision role.

Participant decision and evidence:

CASE 5

SepsisLab — Why clinicians ignore alerts—and how design can restore trust

MANAGEMENT DILEMMA

Should a hospital add another alert, redesign the existing decision-support interaction, or pause deployment until information needs and workflow timing are understood?

Case context

SepsisLab explored why clinicians ignore or distrust sepsis alerts and how a prototype with trends, uncertainty and what-if exploration could support a more collaborative interaction.

Case narrative

The study engaged six experienced US clinicians in discussion of the Epic Sepsis Model and a prototype. Participants considered ranked information needs, trend and uncertainty visualisation, and what-if exploration.

Clinicians described the redesigned prototype more as a collaborator than a static score. This is a qualitative design insight from a very small sample, not evidence of improved clinical outcomes or safety.

The case highlights that an alert can fail even when its score is statistically useful. If it arrives after the decision, hides uncertainty, interrupts without actionability, or generates repeated false positives, clinicians learn to override it.

A patient-centric alert should help before the decision, make change and uncertainty visible, and clarify what information or professional action could reduce risk.

Key facts

- 2024 qualitative study and human-AI prototype evaluation.
- Six experienced US clinicians.
- Trend, uncertainty, ranked information needs and what-if simulation examined.
- Collaborator framing emerged from participant feedback.
- Not an outcome, safety or prospective effectiveness trial.

Primary source: <https://pmc.ncbi.nlm.nih.gov/articles/PMC11149368/>

Case 5 evidence and patient-centric analysis

Reported signal	Evidence label	Limitation	Human checkpoint
Clinicians described the redesigned system more as a collaborator than a static score.	2024 qualitative study and human-AI prototype evaluation	Very small qualitative sample and prototype setting; not an outcome or safety trial.	Clinician interprets risk, orders information and decides action.

Stakeholder map

- Patients at risk of sepsis
- Nurses, physicians and rapid-response teams
- EHR/CDS, quality and patient-safety teams
- Data scientists and vendors
- Leadership responsible for alert burden and outcomes

Patient-centric risks

- False negatives may delay recognition; false positives may create fatigue and unnecessary action.
- Late alerts explain a decision after it was already required.
- A single score may hide trajectory, uncertainty and missing information.
- Unowned alerts can shift responsibility without changing care.

CENTRAL LESSON

Timely, explainable and actionable support can beat another interruptive alert.

Case 5 teaching and transfer guide

Discussion question	Suggested direction
Does the alert arrive before the decision?	Begin with the decision and information need, not the existing score interface.
What changed and why did the alert appear now?	Show trend, uncertainty, contributing data and missing information.
What information could reduce uncertainty?	Provide an owned escalation and documented override route.
Who must review, override, escalate and document?	Measure timeliness, false-positive/negative burden, response, workload, equity and patient outcomes prospectively.
Which prospective measures and stop criteria are required?	Do not claim clinical benefit from a usability prototype alone.

Workshop bridge

Demo	Tools / prompt	Expected output	Human checkpoint
Redesign an unhelpful alert	ChatGPT Prompt 5 • Prompt 5	Before/after alert specification and prospective pilot measures.	Clinical governance approves intended use, thresholds, escalation and monitoring.

- Analyse timeliness, burden, uncertainty, explanation and actionability.
- Specify what changed and why the alert appeared.
- List information that could reduce uncertainty.
- Create override, escalation and audit paths.
- Define pilot measures and stop criteria.

TRANSFER ASSIGNMENT

Bring one non-identifiable alert from your setting and redesign its decision timing, uncertainty and ownership.

Participant decision and evidence:

CASE 6

Ambient scribes — The real workflow cost and payoff of ambient documentation

MANAGEMENT DILEMMA

Should a five-practice group roll out an ambient scribe broadly after promising time-saving reports, or use a controlled pilot that treats review quality as a new clinical process?

Case context

Ambient documentation tools can return attention to the consultation, but they also create a new workflow for consent, review, omission detection, correction, retention and audit.

Case narrative

The supplied implementation dissertation describes a multi-practice deployment with initial resistance, new review work and potential benefits. The programme triangulates this with growing multi-site peer-reviewed observational evidence on ambient documentation.

Across products and settings, reports suggest potential reductions in after-hours EHR work and documentation burden. Outcomes, adoption maturity, measures and product behaviour vary, so local safety and effectiveness cannot be assumed.

Ambient capture changes the consultation: patients need an appropriate notice or consent process, clinicians must review drafts, and the organisation must decide retention, data location, access and incident handling.

The key management insight is that time saved at the keyboard creates a new quality-assurance workflow. A pilot must measure both the returned attention and the errors or omissions that reviewers must catch.

Key facts

- Implementation dissertation plus growing observational health-system evidence.
- Potential reductions in after-hours documentation and burden are reported.
- Product, setting, adoption maturity and outcome measures vary.
- Consent, review, retention and audit are workflow requirements.
- Time saving alone does not establish safety.

Primary source: <https://www.proquest.com/openview/574eebe7a1f1dcd2ca5d88f6a9a22a7b/1?pq-origsite=gscholar&cbl=18750&diss=y>

Case 6 evidence and patient-centric analysis

Reported signal	Evidence label	Limitation	Human checkpoint
Reports describe initial resistance, new review work and potential reductions in after-hours EHR time and burden after adoption.	Implementation dissertation plus growing health-system observational evidence	Product, setting, adoption maturity and outcome measures vary; product-specific safety and local effect require validation.	Clinician or designated reviewer edits and signs; organisation owns consent, retention and audit.

Stakeholder map

- Patients and caregivers in recorded encounters
- Clinicians and designated reviewers
- Health-information management and quality teams
- Privacy, IT/security, procurement and legal teams
- Ambient-scribe vendors and implementation partners

Patient-centric risks

- Patients may not understand capture, retention or use.
- The draft may omit uncertainty, context, negation or sensitive discussion.
- Reviewers may become automation-biased or rush sign-off.
- Data handling may conflict with policy, contract or patient expectation.

CENTRAL LESSON

Time saved at the keyboard creates a new quality-assurance workflow.

Case 6 teaching and transfer guide

Discussion question	Suggested direction
What attention does the tool return to the patient?	Pilot in a bounded service with explicit exclusions and trained reviewers.
What new review work and error types appear?	Use synthetic data before any live encounter and obtain governance approval.
Who reviews the reviewer and how are omissions sampled?	Sample notes using an error taxonomy that includes hallucination, omission, negation, medicine and follow-up errors.
What notice/consent and data-retention model is appropriate?	Measure clinician time, after-hours work, patient experience, correction burden and safety incidents.
Which error would immediately stop the pilot?	Publish rollback and stop criteria; make the final signer accountable.

Workshop bridge

Demo	Tools / prompt	Expected output	Human checkpoint
Ambient scribe versus structured ChatGPT draft	Heidi, Nabla or Abridge demonstration plus ChatGPT Prompts 1 and 6 • Prompt 1	Reviewed draft comparison and controlled-pilot charter.	Clinician/designated reviewer edits and signs; governance owners approve consent, data and audit.

- Label and enter the supplied synthetic consultation.
- Generate a note and patient summary in the specialised tool.
- Run Prompt 1 in ChatGPT with the same transcript.
- Compare omissions, inventions, teach-back and escalation.
- Use Prompt 6 to create a six-week pilot and stop rules.

TRANSFER ASSIGNMENT

Create an ambient-scribe error taxonomy and a weekly sampling plan for a six-week pilot.

Participant decision and evidence:

CASE 7

Explainable AI — Why a visible explanation is necessary—but not sufficient

MANAGEMENT DILEMMA

Should a clinical or pharma team accept a medical-AI product because its predictions are accompanied by attractive explanations?

Case context

Explainable AI methods such as heatmaps, attention maps and feature attribution can support scrutiny, but an explanation may be plausible, unstable or clinically irrelevant.

Case narrative

The 2025 reference chapter discusses explainable-AI applications across imaging, drug discovery and personalised medicine. It illustrates how explanations may support audit, teaching and clinician sense-checking.

As a narrative reference chapter, it is useful for concepts and examples but does not prove that any particular explanation method improves accuracy, safety or patient outcomes.

A heatmap can focus attention while still being unstable, sensitive to model changes or unrelated to clinically meaningful evidence. Explanations should therefore be evaluated alongside intended use, validation, subgroup performance and error consequences.

For patient centricity, explanations must be appropriate to the audience. A technical attribution plot may support professional audit, while a patient needs an honest explanation of what the system did, its uncertainty and who made the final decision.

Key facts

- 2025 narrative reference chapter.
- Covers imaging, drug discovery and personalised medicine examples.
- Discusses heatmaps, attention maps and feature attribution.
- Supports explanation and audit concepts.
- Explanation is not evidence of accuracy, benefit, fairness or regulatory approval.

Primary source: <https://www.sciencedirect.com/science/article/pii/B9780443239793000063>

Case 7 evidence and patient-centric analysis

Reported signal	Evidence label	Limitation	Human checkpoint
The chapter illustrates how explanation can support audit, teaching and clinician sense-checking.	Narrative reference chapter with illustrative case applications	An explanation can be plausible yet unstable or misleading; explainability is not proof of accuracy or benefit.	Professional compares the explanation with intended use, validation and patient context.

Stakeholder map

- Patients and caregivers
- Clinicians, radiologists, pathologists and pharmacists
- Medical, safety and regulatory professionals
- Data-science and model-risk teams
- Procurement, governance and vendors

Patient-centric risks

- A persuasive visual may create false confidence.
- An explanation may not be stable across similar inputs or model versions.
- Technical explanations may be incomprehensible or misleading to patients.
- Teams may overlook subgroup performance, false negatives or weak external validation.

CENTRAL LESSON

Ask why, for whom and under what validation—not only what the model predicted.

Case 7 teaching and transfer guide

Discussion question	Suggested direction
Is the explanation clinically meaningful or merely decorative?	Evaluate explanation as one part of a due-diligence package, not a substitute for performance evidence.
Was the explanation method tested for stability and usefulness?	Ask for intended use, external/local validation, subgroup performance and clinically meaningful error analysis.
What validation population, comparator and subgroup results are available?	Test stability on similar synthetic examples and compare with expert reasoning.
How are false positives and false negatives handled?	Separate professional audit explanations from patient-facing communication.
What explanation does a patient need—and who owns the final decision?	Document override, audit, data and regulatory-status questions before a pilot.

Workshop bridge

Demo	Tools / prompt	Expected output	Human checkpoint
Explainable-AI vendor evaluation	ChatGPT Prompt 8 plus vendor product and validation materials • Prompt 8	Auditable vendor-question list and pilot/no-pilot rationale.	Clinical, regulatory, privacy, security and procurement reviewers verify original evidence and current status.

- Extract intended use, user, population and setting.
- Map validation, comparator, false positives/negatives and subgroups.
- Describe the explanation method and its claimed clinical meaning.
- Identify evidence gaps, India applicability and data controls.
- State reasons to pilot and reasons not to pilot.

TRANSFER ASSIGNMENT

Review one medical-AI product page and mark each statement as evidence, marketing claim, unknown or decision requirement.

Participant decision and evidence:

Part II — 90-minute demonstration workshop

WORKSHOP PROMISE

Every specialised-tool demonstration is paired with a ChatGPT exercise. The learning objective survives a paid account, regional restriction or network failure because the same synthetic case, comparison and human checkpoint remain.

Clock	Activity	Minutes
00–08	Launch, access map and universal wrapper	8
08–26	Station 1 — Consultation continuity	18
26–49	Station 2 — Diagnostic support + medicines	23
49–68	Station 3 — Patient + pharma communication	19
68–83	Station 4 — Research + teaching	15
83–90	POPLE scorecard, 30-day capstone and close	7
TOTAL	Hands-on laboratory	90

STATION 1		Consultation continuity — 18 minutes
Specialised tools	ChatGPT prompts	Participant output
Heidi Health • Nabla • Abridge • ChatGPT	P1, P6	Reviewed draft encounter summary, patient explanation, teach-back questions and controlled ambient-scribe pilot plan.

SYNTHETIC / APPROVED INPUT

Mrs A, 52: type 2 diabetes and hypertension; missed medicines twice this week; low health literacy; prefers Hindi; tingling feet; believes insulin means failure. No acute red flags are supplied.

Team roles

- Facilitator/tool operator
- Clinician reviewer
- Patient advocate
- Evidence/privacy observer
- Reporter

Facilitator demonstration sequence

1. Label the case synthetic and explain the consent/notice script.
2. Enter the prepared transcript into the approved ambient tool or use saved screenshots.
3. Generate the draft note and patient summary.
4. Run ChatGPT Prompt 1 with the same input.
5. Compare facts, omissions, inventions, medicine questions and escalation.
6. Edit, approve or reject; then use Prompt 6 for pilot governance.

Station 1 verification and debrief

Human checkpoint	Stop rule	Fallback
No content enters a record or reaches a patient until a clinician/designated reviewer corrects and signs it.	Stop if an invented diagnosis, medicine, dose, allergy or red-flag disposition passes review unnoticed.	Use prepared Heidi/Nabla/Abridge screenshots or saved output, then run Prompt 1 and preserve the same comparison and debrief.

DEBRIEF QUESTION

What attention did the tool return—and what new verification work did it create?

Comparison	Specialised tool	ChatGPT	Final human decision
Facts / inventions			
Urgency / escalation			
Evidence / traceability			
Privacy / consent			
Patient value			
Workload / feasibility			

Approved participant output:

STATION 2

Diagnostic support + medication safety — 23 minutes

Specialised tools	ChatGPT prompts	Participant output
DxGPT • Medscape AI • ClinicalKey AI • VisualDx • iatroX • MDCalc • Epocrates • ChatGPT	P2, P3	Structured uncertainty map, source-verification record and medicine-reconciliation checklist—not a diagnosis or prescription.

SYNTHETIC / APPROVED INPUT

Mr R, 58: progressive breathlessness and ankle swelling; BP 164/96, pulse 104, SpO₂ 93%; diabetes, hypertension and smoking history; raised BNP and basal crepitations; medicine and ECG information incomplete.

Team roles

- Facilitator/tool operator
- Clinical reasoner
- Urgency checker
- Evidence checker
- Pharmacist/prescriber role

Facilitator demonstration sequence

1. Ask which information changes urgency before it changes the differential.
2. Run the problem representation in DxGPT or show prepared specialised-tool output.
3. Run ChatGPT Prompt 2 with exactly the same synthetic facts.
4. Check one claim in Medscape AI/ClinicalKey AI and open the cited source.
5. Demonstrate one bounded MDCalc input or medicine-reference verification.
6. Record agreement, conflict, local-protocol fit and the clinician-owned decision.

Station 2 verification and debrief

Human checkpoint	Stop rule	Fallback
A licensed clinician owns urgency, diagnosis and disposition; a prescriber/pharmacist verifies medicines and approved sources.	Stop if the workflow delays urgent escalation, invents a finding or turns a hypothesis into a diagnosis.	Use saved specialised outputs and one opened guideline/monograph; participants still run Prompts 2–3 and triangulate.

DEBRIEF QUESTION

Did the tools widen attention or create noise? Which claim remained unverified?

Comparison	Specialised tool	ChatGPT	Final human decision
Facts / inventions			
Urgency / escalation			
Evidence / traceability			
Privacy / consent			
Patient value			
Workload / feasibility			

Approved participant output:

STATION 3

Patient + pharma communication — 19 minutes

Specialised tools	ChatGPT prompts	Participant output
Canva • Gamma • Bhashini • approved MI/PV workflow • ChatGPT	P4, P7, P10	Reviewed patient asset, MI/PV/PQC/MLR routing plan and ethical professional-projection draft.

SYNTHETIC / APPROVED INPUT

Approved public source excerpt for a class 6–8 patient handout, plus a separate synthetic unsolicited medical-information interaction containing a possible adverse event and an off-label question.

Team roles

- Clinical content reviewer
- Patient/literacy advocate
- Bilingual reviewer
- Medical-information/MLR role
- PV/PQC escalation role

Facilitator demonstration sequence

1. Use Prompt 4 to draft a one-page patient handout from the approved source only.
2. Review meaning, red flags, teach-back and unsupported [REVIEW] statements.
3. Use Canva/Gamma for design and Bhashini only as a draft language aid.
4. Run Prompt 7 on the synthetic pharma inquiry and separate MI, off-label, PV/PQC and MLR responsibilities.
5. Use Prompt 10 to create a useful, evidence-bounded professional asset.
6. Apply the named clinical, bilingual and regulatory release gates.

Station 3 verification and debrief

Human checkpoint	Stop rule	Fallback
Nothing reaches a patient or external audience until the appropriate clinical, bilingual, MI/PV and MLR reviewers approve it.	Stop for unsupported claims, mistranslated clinical meaning, promotional drift or delayed adverse-event/product-quality escalation.	Create a paper wireframe from Prompt 4; role-play the MI/PV interaction with Prompt 7; keep the same release gates.

DEBRIEF QUESTION

Can the patient act safely, and is every professional boundary visible?

Comparison	Specialised tool	ChatGPT	Final human decision
Facts / inventions			
Urgency / escalation			
Evidence / traceability			
Privacy / consent			
Patient value			
Workload / feasibility			

Approved participant output:

STATION 4

Research + Socratic teaching — 15 minutes

Specialised tools	ChatGPT prompts	Participant output
PubMed • Elicit • ResearchRabbit • Scite • ChatGPT	P11, P9	Transparent starting evidence map, uncertainty log and staged Socratic teaching script—not a fabricated systematic review.

SYNTHETIC / APPROVED INPUT

PICO: ambient documentation versus usual documentation; outcomes include after-hours EHR time, note errors and patient experience. Use one verified seed source.

Team roles

- Question/PICO lead
- Search recorder
- Study-design checker
- Citation verifier
- Socratic facilitator

Facilitator demonstration sequence

1. Write the PICO and the decision it will inform.
2. Build and record a PubMed query, date, filters and result count.
3. Use Elicit to structure candidate evidence and verify fields in original records.
4. Use ResearchRabbit for adjacency and Scite for citation context—not quality verdicts.
5. Run Prompt 11 to create an evidence-map template and uncertainty log.
6. Run Prompt 9 to create a commit–challenge–verify–revise dialogue.

Station 4 verification and debrief

Human checkpoint	Stop rule	Fallback
Open every source; verify design, population, outcome and limitation; faculty protects learner reasoning and originality.	Stop when a citation cannot be opened, a design is mislabelled or an AI summary outruns the source.	Use saved PubMed records, citation maps and citation-context screenshots; preserve verification and appraisal.

DEBRIEF QUESTION

What is the defensible deliverable—and what remains outside this rapid workflow?

Comparison	Specialised tool	ChatGPT	Final human decision
Facts / inventions			
Urgency / escalation			
Evidence / traceability			
Privacy / consent			
Patient value			
Workload / feasibility			

Approved participant output:

Part III — Eleven copy-ready ChatGPT prompts

PROMPT-USE METHOD

Paste the universal wrapper first. Replace only bracketed placeholders with synthetic/approved information. Compare output with the specialised tool where shown, then preserve the human-verification section.

Prompt	Purpose	Specialised-tool comparison
P1	Consultation continuity	ChatGPT; compare with Heidi or Nabla
P2	Diagnostic reasoning without outsourcing diagnosis	ChatGPT; compare with DxGPT and a clinical evidence tool
P3	Medication-reconciliation discussion	ChatGPT; compare with Epocrates, Lexicomp, Micromedex or an approved reference
P4	Patient-education asset	ChatGPT plus Canva/Gamma; bilingual review may use Bhashini
P5	Redesigning an unhelpful clinical alert	ChatGPT discussion prompt
P6	Ambient-scribe pilot plan	ChatGPT; compare with Heidi/Nabla governance settings
P7	Pharma medical-information and safety triage	ChatGPT discussion prompt; compare with approved medical-information/PV workflow
P8	Explainable-AI vendor evaluation	ChatGPT discussion prompt
P9	Socratic medical teaching	ChatGPT
P10	Professional projection	ChatGPT plus approved publishing/design workflow
P11	Rapid evidence map	ChatGPT plus PubMed/Elicit/ResearchRabbit/Scite

PROMPT 1

Consultation continuity

Turn a synthetic conversation into a reviewable draft without inventing clinical facts.

You are supporting a synthetic healthcare education exercise. You are not the treating clinician and must not replace professional judgement. Do not provide a final diagnosis, prescribe, change a dose, provide reassurance or instruct anyone to delay urgent care. Separate supplied facts, patient-reported information, professional observations, assumptions, unknowns and AI-generated hypotheses. Do not invent findings, medicines, laboratory results, citations or guidelines. Identify red flags and information requiring urgent professional review. Use only the information and approved sources supplied. End with a section titled 'Human decision and verification required'.

TASK

Using the synthetic consultation transcript below, create: (1) a source-grounded draft encounter summary, (2) a problem-information list, (3) medication-reconciliation questions, (4) a plain-language patient summary, (5) three teach-back questions, and (6) follow-up and escalation items. Tag every statement [PATIENT REPORTED], [PROFESSIONAL OBSERVATION], [UNKNOWN] or [NEEDS VERIFICATION]. Do not invent examination findings, diagnosis, medication, dose, investigation, reassurance or follow-up decision. Transcript: [PASTE SYNTHETIC TRANSCRIPT]

Expected output	Human checkpoint	Failure mode
Tagged draft note, patient summary, questions and escalation items	Reconcile facts and medicines; check red flags, consent, local language and record policy before sign-off.	Invented findings or a polished draft copied into the record without review.

DEBRIEF QUESTION

What attention did the tool return—and what new verification work did it create?

PROMPT 2

Diagnostic reasoning without outsourcing diagnosis

Expose hypotheses, missing information and urgency.

You are supporting a synthetic healthcare education exercise. You are not the treating clinician and must not replace professional judgement. Do not provide a final diagnosis, prescribe, change a dose, provide reassurance or instruct anyone to delay urgent care. Separate supplied facts, patient-reported information, professional observations, assumptions, unknowns and AI-generated hypotheses. Do not invent findings, medicines, laboratory results, citations or guidelines. Identify red flags and information requiring urgent professional review. Use only the information and approved sources supplied. End with a section titled 'Human decision and verification required'.

TASK

Using only the synthetic case below: (1) write a one-sentence problem representation without declaring a diagnosis; (2) group hypotheses as common/likely, dangerous must-not-miss, and alternative explanations; (3) for each, list supporting information, contradictory information, missing information, red flags, time sensitivity and evidence that must be verified; (4) identify the five questions or observations most likely to change urgency; and (5) state what a licensed clinician must decide. Do not recommend treatment or invent probabilities. Case: [PASTE SYNTHETIC CASE]

Expected output	Human checkpoint	Failure mode
Structured hypotheses and uncertainty map	Check urgent red flags, primary evidence, local protocol and patient context; clinician owns diagnosis and disposition.	A plausible list mistaken for a diagnosis or a reason to delay escalation.

DEBRIEF QUESTION

Did the output widen attention or create noise?

PROMPT 3

Medication-reconciliation discussion

Surface possible medicine issues that require verification.

You are supporting a synthetic healthcare education exercise. You are not the treating clinician and must not replace professional judgement. Do not provide a final diagnosis, prescribe, change a dose, provide reassurance or instruct anyone to delay urgent care. Separate supplied facts, patient-reported information, professional observations, assumptions, unknowns and AI-generated hypotheses. Do not invent findings, medicines, laboratory results, citations or guidelines. Identify red flags and information requiring urgent professional review. Use only the information and approved sources supplied. End with a section titled 'Human decision and verification required'.

TASK

Review the synthetic medication information below. Do not recommend starting, stopping or changing any medicine. Create a verification checklist covering possible duplication, interaction, allergy conflict, renal/hepatic considerations, pregnancy/lactation if relevant, indication uncertainty, adherence, OTC/supplement/traditional-medicine use, monitoring and missing name/strength/formulation/route/frequency. Separate confirmed facts, potential issues to verify and missing information. List questions for the patient and prescriber/pharmacist, plus the approved drug-information sources that must be checked. Information: [PASTE SYNTHETIC LIST]

Expected output	Human checkpoint	Failure mode
Verification checklist and escalation questions	Licensed prescriber/pharmacist checks labels, formulary, renal/hepatic context and local protocol.	Model invents an interaction or silently changes a dose.

DEBRIEF QUESTION

Which issue is a signal—and which is a verified fact?

PROMPT 4

Patient-education asset

Convert approved evidence into actionable plain language.

You are supporting a synthetic healthcare education exercise. You are not the treating clinician and must not replace professional judgement. Do not provide a final diagnosis, prescribe, change a dose, provide reassurance or instruct anyone to delay urgent care. Separate supplied facts, patient-reported information, professional observations, assumptions, unknowns and AI-generated hypotheses. Do not invent findings, medicines, laboratory results, citations or guidelines. Identify red flags and information requiring urgent professional review. Use only the information and approved sources supplied. End with a section titled 'Human decision and verification required'.

TASK

Using only the approved source excerpt below, draft a one-page patient handout for [AUDIENCE] at class 6–8 reading level. Include a short title, five practical actions, medicine-taking reminders without products or doses, monitoring, routine follow-up, three urgent red flags, three teach-back questions, source, review date, reviewer placeholder and AI-assistance disclosure. Mark unsupported statements [REVIEW]. Provide a draft Hindi version, marking medical terms for bilingual professional review. Do not invent statistics, guarantees or clinical advice outside the source. Source: [PASTE APPROVED SOURCE]

Expected output	Human checkpoint	Failure mode
Plain-language handout with red flags and teach-back	Clinical meaning, translation, accessibility, source, version and reviewer approval.	Unsafe simplification, mistranslation or an unsupported promise.

DEBRIEF QUESTION

Can the patient act safely—and is routine advice separated from urgent action?

PROMPT 5

Redesigning an unhelpful clinical alert

Apply the SepsisLab design lesson to a synthetic alert.

You are supporting a synthetic healthcare education exercise. You are not the treating clinician and must not replace professional judgement. Do not provide a final diagnosis, prescribe, change a dose, provide reassurance or instruct anyone to delay urgent care. Separate supplied facts, patient-reported information, professional observations, assumptions, unknowns and AI-generated hypotheses. Do not invent findings, medicines, laboratory results, citations or guidelines. Identify red flags and information requiring urgent professional review. Use only the information and approved sources supplied. End with a section titled 'Human decision and verification required'.

TASK

Evaluate the synthetic alert below without making a patient-specific treatment recommendation. Analyse timeliness, false-positive burden, false-negative risk, explanation, uncertainty, actionability, workflow interruption, override, escalation and audit trail. Redesign it to show what changed, why it appeared, current uncertainty, information that could reduce uncertainty, required professional review, override/documentation path, pilot metrics and stop criteria. Alert: [PASTE SYNTHETIC ALERT]

Expected output	Human checkpoint	Failure mode
Before/after alert specification and pilot measures	Clinical governance approves intended use, escalation, thresholds and monitoring.	A more attractive alert that still arrives late or triggers unowned action.

DEBRIEF QUESTION

Does the redesign help before the decision—or simply explain after it?

PROMPT 6

Ambient-scribe pilot plan

Design a controlled implementation with quality assurance.

You are supporting a synthetic healthcare education exercise. You are not the treating clinician and must not replace professional judgement. Do not provide a final diagnosis, prescribe, change a dose, provide reassurance or instruct anyone to delay urgent care. Separate supplied facts, patient-reported information, professional observations, assumptions, unknowns and AI-generated hypotheses. Do not invent findings, medicines, laboratory results, citations or guidelines. Identify red flags and information requiring urgent professional review. Use only the information and approved sources supplied. End with a section titled 'Human decision and verification required'.

TASK

Design a six-week controlled ambient-scribe pilot. Include intended and excluded uses; patient notice/consent; minimum necessary data; clinician and scribe-reviewer responsibilities; note sampling; error taxonomy; hallucination and omission monitoring; patient-experience, clinician-time and after-hours documentation measures; safety escalation; rollback; stop criteria; and final continue/redesign/stop decision. Do not assume time saving proves safety.

Expected output	Human checkpoint	Failure mode
Six-week pilot charter and measurement plan	Privacy, clinical governance, IT/security and accountable service owner approve the pilot.	Plug-and-play rollout without consent, sampling or stop rules.

DEBRIEF QUESTION

Who reviews the reviewer—and what error stops the pilot?

PROMPT 7

Pharma medical-information and safety triage

Separate scientific response, off-label handling and immediate safety escalation.

You are supporting a synthetic healthcare education exercise. You are not the treating clinician and must not replace professional judgement. Do not provide a final diagnosis, prescribe, change a dose, provide reassurance or instruct anyone to delay urgent care. Separate supplied facts, patient-reported information, professional observations, assumptions, unknowns and AI-generated hypotheses. Do not invent findings, medicines, laboratory results, citations or guidelines. Identify red flags and information requiring urgent professional review. Use only the information and approved sources supplied. End with a section titled 'Human decision and verification required'.

TASK

Analyse the synthetic medical-information interaction. Create: the exact unsolicited scientific question; minimum information needed; approved sources; a neutral response outline; statements requiring medical/legal/regulatory review; possible off-label elements requiring medical-information handling; possible adverse-event or product-quality information requiring immediate escalation; follow-up commitments; and documentation requirements. Do not create promotional claims, assess causality, minimise an event or delay escalation. Interaction: [PASTE SYNTHETIC INTERACTION]

Expected output	Human checkpoint	Failure mode
Neutral inquiry plan plus explicit safety/PQC escalation flags	Medical information, pharmacovigilance and MLR follow the approved SOP; safety escalation is immediate.	A polished response that misses or delays an adverse-event report.

DEBRIEF QUESTION

Which part belongs to medical information, PV, product quality or MLR?

PROMPT 8

Explainable-AI vendor evaluation

Turn marketing claims into an auditable due-diligence conversation.

You are supporting a synthetic healthcare education exercise. You are not the treating clinician and must not replace professional judgement. Do not provide a final diagnosis, prescribe, change a dose, provide reassurance or instruct anyone to delay urgent care. Separate supplied facts, patient-reported information, professional observations, assumptions, unknowns and AI-generated hypotheses. Do not invent findings, medicines, laboratory results, citations or guidelines. Identify red flags and information requiring urgent professional review. Use only the information and approved sources supplied. End with a section titled 'Human decision and verification required'.

TASK

Evaluate the medical-AI product description. Build a due-diligence table for intended use/user, target population, setting, validation population and comparator, sensitivity/specificity, false positives/negatives, subgroup performance, external validation, explainability method and clinical meaning, workflow integration, override, audit trail, data location/retention, jurisdiction-specific regulatory status, evidence gaps, India applicability, vendor questions, reasons to pilot and reasons not to pilot. Do not infer approval or effectiveness from marketing language. Description: [PASTE]

Expected output	Human checkpoint	Failure mode
Vendor-question list and pilot/no-pilot rationale	Clinical, regulatory, privacy, security and procurement reviewers verify original evidence and status.	Decorative heatmaps treated as proof of correctness.

DEBRIEF QUESTION

Is the explanation clinically meaningful or merely persuasive?

PROMPT 9

Socratic medical teaching

Use AI to strengthen reasoning rather than reveal an answer immediately.

You are supporting a synthetic healthcare education exercise. You are not the treating clinician and must not replace professional judgement. Do not provide a final diagnosis, prescribe, change a dose, provide reassurance or instruct anyone to delay urgent care. Separate supplied facts, patient-reported information, professional observations, assumptions, unknowns and AI-generated hypotheses. Do not invent findings, medicines, laboratory results, citations or guidelines. Identify red flags and information requiring urgent professional review. Use only the information and approved sources supplied. End with a section titled 'Human decision and verification required'.

TASK

Act as a Socratic facilitator for the synthetic case. Do not reveal the answer immediately. Ask the learner to create a problem representation; propose common, dangerous and alternative hypotheses; identify evidence for/against; identify missing information and urgency; challenge one assumption; introduce one verified source; ask the learner to revise; then provide feedback with a transparent rubric and identify unsupported AI-generated information. Case: [PASTE]

Expected output	Human checkpoint	Failure mode
Staged questions, revised reasoning and rubric feedback	Faculty verifies content and preserves learner effort, dialogue and originality.	Instant answers reduce productive struggle and create look-alike submissions.

DEBRIEF QUESTION

Did the AI promote reasoning or replace it?

PROMPT 10

Professional projection

Draft a credible evidence-based professional post with visible boundaries.

You are supporting a synthetic healthcare education exercise. You are not the treating clinician and must not replace professional judgement. Do not provide a final diagnosis, prescribe, change a dose, provide reassurance or instruct anyone to delay urgent care. Separate supplied facts, patient-reported information, professional observations, assumptions, unknowns and AI-generated hypotheses. Do not invent findings, medicines, laboratory results, citations or guidelines. Identify red flags and information requiring urgent professional review. Use only the information and approved sources supplied. End with a section titled 'Human decision and verification required'.

TASK

Using only the approved evidence below, draft a 150–200 word professional post for [DOCTOR / MEDICAL-AFFAIRS PROFESSIONAL] to [EDUCATE PATIENTS / EDUCATE PEERS / DISCUSS AN EVIDENCE GAP]. Include one evidence-based message, one practical implication, one limitation, a visible source, AI-assistance disclosure and an invitation to professional discussion—not individual treatment advice. Do not add product promotion, off-label claims, exaggerated certainty or authority beyond the stated role. Evidence: [PASTE]

Expected output	Human checkpoint	Failure mode
Concise, sourced and bounded professional post	Clinical/MLR review, citation, disclosure, version owner and publication approval.	Authority signalling, promotion or advice without patient context.

DEBRIEF QUESTION

Does the post prove usefulness before claiming authority?

PROMPT 11

Rapid evidence map

Create a reproducible starting map without fabricating citations.

You are supporting a synthetic healthcare education exercise. You are not the treating clinician and must not replace professional judgement. Do not provide a final diagnosis, prescribe, change a dose, provide reassurance or instruct anyone to delay urgent care. Separate supplied facts, patient-reported information, professional observations, assumptions, unknowns and AI-generated hypotheses. Do not invent findings, medicines, laboratory results, citations or guidelines. Identify red flags and information requiring urgent professional review. Use only the information and approved sources supplied. End with a section titled 'Human decision and verification required'.

TASK

Convert the question into a transparent rapid-research workflow. Produce PICO; concepts and synonyms; a draft PubMed Boolean query; inclusion and exclusion criteria; evidence-extraction fields; citation-verification checklist; bias/applicability questions; uncertainty log; and a statement of what this rapid review cannot conclude. Do not invent citations. Label every citation [UNVERIFIED] until located in PubMed, a registry or the original publication. Question: [PASTE]

Expected output	Human checkpoint	Failure mode
Search strategy, extraction frame and uncertainty log	Every citation and extracted field is verified in the original source; methods and limitations are documented.	A fluent synthesis with fabricated or mischaracterised studies.

DEBRIEF QUESTION

Which evidence was actually verified—and why is this not a systematic review?

Part IV — Medical tool and implementation toolkit

Tool	Category	Access	India note	Official/demo URL	Privacy / data checkpoint
Heidi Health	Ambient documentation	Free tier + paid	Accessible; confirm institutional approval	https://scribe.heidihealth.com/	Consent, retention, hosting and institutional DPA must be checked before clinical use
Nabla	Ambient documentation	Free trial / paid enterprise	Web access possible; confirm regional contracting and data location	https://app.nabla.com/	Requires consent, DPA, retention and EHR governance review
Abridge	Ambient documentation	Paid enterprise / request demo	No public India self-serve path identified	https://www.abridge.com/product	Enterprise security, EHR integration and contracting required
Medscape AI	Clinical evidence	Free for registered users	Generally accessible to registered users; confirm account access	https://www.medscape.com/sites/public/medscape-ai	Use de-identified questions; do not treat as patient record system
ClinicalKey AI	Clinical evidence	Paid institutional / request demo	Available through institutional procurement	https://www.elsevier.com/products/clinicalkey/clinicalkey-ai/request-demo	Institutional access and governance; avoid identifiable patient data unless approved
VisualDx	Clinical decision support	Paid + trial / institutional	Accessible; institutional access may be required	https://www.visualdx.com/	Images require consent, de-identification and organisational controls
DxGPT	Differential diagnosis	Free	Accessible globally	https://dxgpt.app/	Vendor states encryption and anonymisation; still avoid identifiable workshop data
iatroX	Clinical learning and reasoning	Free; verify current scope	Accessible	https://iatrox.com/	Use de-identified or synthetic cases
MDCalc	Clinical calculators	Free core access	Accessible	https://www.mdcalc.com/	Do not store identifiers; verify inputs and applicability
Epocrates	Drug information	Free core + paid premium	Accessible; content may be US-oriented	https://www.epocrates.com/	Reference use; verify local brands, approvals and protocols
ChatGPT	General AI assistant	Free tier + paid	Accessible	https://chatgpt.com/	Never paste identifiable clinical data into public/general accounts; use approved enterprise controls
Canva Magic Studio	Professional communication	Free tier + paid	Accessible	https://www.canva.com/magic-studio/	Use approved assets; no patient images/data without consent; review generated medical content
Gamma	Professional communication	Free tier + paid	Accessible	https://gamma.app/	Do not upload confidential or identifiable data; fact-check medical content

Tool	Category	Access	India note	Official/demo URL	Privacy / data checkpoint
Bhashini	Language enablement	Government platform / access varies	India-specific	https://bhashini.gov.in/ulca/model/explore-models	Do not send clinical identifiers without approved architecture
PubMed	Literature search	Free	Accessible	https://learn.nlm.nih.gov/documentation/training-packets/T0042010P/	Search tool; do not paste identifiable clinical narratives
Elicit	Research synthesis	Free tier + paid	Accessible	https://elicit.com/	Do not upload confidential manuscripts or patient data without approval
ResearchRabbit	Research mapping	Free core access	Accessible	https://www.researchrabbit.ai/	Avoid confidential documents unless terms permit
Scite	Citation context	Limited free access + paid	Accessible	https://scite.ai/	Avoid confidential uploads unless approved

ACCESS WARNING

Free, free-tier, paid and institutional labels describe the checked public offer; geography, verification, features, pricing and status may change. Reopen every official link before delivery.

POPLE tool scorecard

Dimension	Question	Score 1–5	Evidence
Patient value	Attention, understanding, access, continuity or safety?		
Evidence	Can original sources and validation limits be opened?		
Ownership	Who reviews, releases, escalates and audits?		
Privacy	Is data minimised, consented, secured and retained appropriately?		
Local fit	Does it fit India access, language, protocol and workflow?		
Effort	What training, review burden, integration and cost appear?		

AUTOMATIC GATE

Do not advance a workflow with Ownership or Privacy below 3/5, even when its overall score is high.

Demo readiness checklist

Check	Ready?	Evidence / owner
Official URL and demo link open	☐	
Account, profession and region verified	☐	
Privacy/institutional approval confirmed	☐	
Synthetic input visibly labelled	☐	
Expected output and human checkpoint known	☐	
Screenshot/saved-output fallback opens	☐	
Stop rule and urgent escalation announced	☐	
Timer, group roles and debrief protected	☐	

Part V — Assessment and 30-day transfer

Knowledge check

1. What is the first question before selecting an AI tool?

2. What evidence design is the Qatar case?

3. What is the safe role of an AI differential?

4. What should overlap before accepting a clinical claim?

5. What new workflow does an ambient scribe create?

6. What does an explainability heatmap prove?

7. What does rapid research honestly deliver?

8. What pharma information requires immediate escalation?

9. Which POPLD dimensions are minimum gates?

10. What happens when a live tool fails?

Answer key

#	Question	Expected answer
1	What is the first question before selecting an AI tool?	Will this improve patient attention, understanding, safety or continuity?
2	What evidence design is the Qatar case?	A cross-sectional perception survey—not an effectiveness trial.
3	What is the safe role of an AI differential?	A hypothesis and missing-information aid; a clinician owns diagnosis/disposition.
4	What should overlap before accepting a clinical claim?	AI output, opened original/approved evidence and local patient/protocol context.
5	What new workflow does an ambient scribe create?	Consent, review, correction, retention, audit and incident handling.
6	What does an explainability heatmap prove?	Nothing by itself about accuracy, benefit, fairness or approval.
7	What does rapid research honestly deliver?	A transparent starting map and uncertainty log—not a completed systematic review.
8	What pharma information requires immediate escalation?	A possible adverse event or product-quality complaint.
9	Which POPLD dimensions are minimum gates?	Ownership and privacy.
10	What happens when a live tool fails?	Use prepared output and the paired prompt; preserve checkpoint and debrief.

My 30-day controlled experiment

Step	Decision	Plan
Use case	One friction point + one patient value	
Baseline	Time, errors, experience, access or quality before AI	
Pilot	Small scope; synthetic/test first; named owner	
Check	Sampling, POPLD, incident path and stop rule	
Decide	Continue, redesign or stop on a named date	

My commitment and accountable reviewer:

References and source register

Seven primary case sources

- Case 1 — EHR-embedded X-ray triage and no-show prediction — <https://www.mdpi.com/2073-431X/14/8/320>
- Case 2 — What family physicians expect and fear — <https://ai.jmir.org/2024/1/e40781>
- Case 3 — Generative AI in case-based medical learning — <https://link.springer.com/article/10.1186/s12909-025-07567-z>
- Case 4 — AI-supported scenario planning for medical curricula — <https://www.jmir.org/2025/1/e73173>
- Case 5 — Why clinicians ignore alerts—and how design can restore trust — <https://pmc.ncbi.nlm.nih.gov/articles/PMC11149368/>
- Case 6 — The real workflow cost and payoff of ambient documentation — <https://www.proquest.com/openview/574eebe7a1f1dcd2ca5d88f6a9a22a7b/1?pq-origsite=gscholar&cbl=18750&diss=y>
- Case 7 — Why a visible explanation is necessary—but not sufficient — <https://www.sciencedirect.com/science/article/pii/B9780443239793000063>

Governance and supporting evidence

- Digital Personal Data Protection Rules, 2025 (MeitY, Government of India, 2025) — <https://www.meity.gov.in/documents/act-and-policies/digital-personal-data-protection-rules-2025-gDOxUjMtQWa?pageTitle=Digital-Personal-Data-Protection-Rules-2025> — Use: Purpose limitation, notice, security and breach governance
- Ethical Guidelines for Application of AI in Biomedical Research and Healthcare (ICMR, 2023) — <https://www.icmr.gov.in/ethical-guidelines-for-application-of-artificial-intelligence-in-biomedical-research-and-healthcare> — Use: Ethics, accountability, transparency, bias, consent and governance
- Telemedicine Practice Guidelines / FAQs (National Medical Commission, 2020) — https://www.nmc.org.in/MCIRest/open/getDocument?path=%2FDocuments%2FPublic%2FPortal%2FLatestNews%2FFinal_FAQ-TELEMEDICINE++6-4-2020..pdf — Use: Consent, identification, documentation and clinical responsibility
- Draft Guidance Document on Medical Device Software (CDSCO, 2025) — https://cdsco.gov.in/opencms/resources/UploadCDSCOWeb/2018/UploadPublic_NoticesFiles/Draft%20guidance%20document%20on%20Medical%20Device%20Software%2021%2010%202025.pdf — Use: Intended use, classification and regulatory pathway for medical device software
- Uniform Code for Pharmaceutical Marketing Practices 2024 (Department of Pharmaceuticals, 2024; amended 2025) — <https://pharma-dept.gov.in/policy/uniform-code-pharmaceutical-marketing-practices-ucpmp-2024-reg> — Use: Promotion, medical communication, transparency and HCP engagement
- Uniform Code for Marketing Practices in Medical Devices 2024 (Department of Pharmaceuticals, 2024; amended 2026) — <https://pharma-dept.gov.in/policy/uniform-code-marketing-practices-medical-devices-ucmpmd-2024> — Use: Medical-device communication and promotion
- eSanjeevani Programme (Online OPD) (Government of India, Current) — <https://www.india.gov.in/category/health-wellness/subcategory/health-resources/details/esanjeevani-programme-online-opd> — Use: India telemedicine service
- Use of Ambient AI Scribes to Reduce Administrative Burden and Improve Clinician Well-being (JAMA Network Open, 2025) — <https://jamanetwork.com/journals/jamanetworkopen/fullarticle/2839542> — Use: Ambient-scribe benefits and limitations
- Impact of an Ambient AI Scribe Among Clinicians and Patients (JMIR Medical Informatics, 2026) — <https://medinform.jmir.org/2026/1/e85580> — Use: Clinician and patient outcomes
- Changes in Clinician Time Expenditure and Visit Quantity Following Ambient AI Scribe Use (JAMA, 2026) — <https://jamanetwork.com/journals/jama/article-abstract/2847319> — Use: Time-use evidence

- Multi-model assurance analysis of clinical hallucination risk (Communications Medicine, 2025) — <https://www.nature.com/articles/s43856-025-01021-3> — Use: Hallucination and adversarial misinformation risk
- Framework to assess clinical safety and hallucination rates (npj Digital Medicine / PMC, 2025) — <https://pmc.ncbi.nlm.nih.gov/articles/PMC12075489/> — Use: Structured safety evaluation
- Artificial Intelligence-Supported Patient-Centered Clinical Decision Support (AHRQ, 2024) — https://digital.ahrq.gov/sites/default/files/IAS%20Topic%20Highlight%20AI%20and%20PC%20CDS_508%20Compliant.pdf — Use: Shared decision-making and patient-centred CDS
- Patient perspectives on clinicians' use of ambient AI scribes (JAMIA Open, 2026) — <https://academic.oup.com/jamiaopen/article/9/3/ooag104/8707573> — Use: Patient acceptability and preference evidence
- Medscape AI (Medscape, Current) — <https://www.medscape.com/sites/public/medscape-ai> — Use: Clinical evidence tool and access
- OpenEvidence (OpenEvidence, Current) — <https://www.openevidence.com/> — Use: US clinician access restriction and evidence search
- ClinicalKey AI (Elsevier, Current) — <https://www.elsevier.com/products/clinicalkey/clinicalkey-ai> — Use: Evidence-grounded clinical decision support
- VisualDx (VisualDx, Current) — <https://www.visualdx.com/> — Use: Visual clinical decision support
- Heidi Health (Heidi Health, Current) — <https://www.heidihealth.com/> — Use: Ambient documentation and free entry point
- Nabla (Nabla, Current) — <https://www.nabla.com/> — Use: Ambient documentation and free trial
- DxGPT FAQ (Foundation 29, Current) — <https://dxgpt.app/faq> — Use: Free access, decision-support status and limitations
- Veeva Vault Medical (Veeva, Current) — <https://www.veeva.com/products/vault-medical/> — Use: Medical-affairs workflow

FINAL PRINCIPLE

AI assists. Professionals decide. Patients remain the purpose.

End of complete handbook